Predictive ability of preinjury stressful life events and post-traumatic stress symptoms for outcomes following mild traumatic brain injury: analysis in a prospective emergency room sample

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Received 27 August 2010
Accepted 8 November 2010
Published Online First 17 January 2011

ABSTRACT

Background A subset of persons with mild traumatic brain injury (mTBI) experience long-term difficulties. Preinjury stress has been hypothesised to play a role in long-term maintenance of symptoms.

Objective To investigate the predictive ability of preinjury stressful life events and post-traumatic stress symptoms to health-related quality of life and emotional distress after mTBI.

Methods Within 2 weeks of injury, 186 participants with mTBI who were admitted to an emergency centre completed an interview and questionnaires regarding preinjury functioning, including the Stressful Life Events Questionnaire and the Post-Traumatic Stress Disorder Checklist. Outcomes were assessed at 3 months after injury and included the depression and anxiety subscales of the Brief Symptom Inventory, and the physical and mental component scores of the 36-item Short-Form Health Survey (SF-36). The incidence and type of stressful life events were reported. Hierarchical regression analyses were used to determine the predictive utility of Stressful Life Events Questionnaire and Post-Traumatic Stress Disorder Checklist after controlling for age, injury severity (complicated versus uncomplicated mild) and preinjury depression.

Results Several potentially life-altering stressful events were endorsed by at least 25% of participants as having been experienced prior to injury. The incidence of stressful life events was a significant predictor of all four outcome variables. History of post-traumatic stress symptoms was predictive of scores on the SF-36 mental health component.

Conclusions A history of stressful events may predispose persons with mTBI to have poor outcomes. History of stress should be assessed during the early stages after mTBI to help identify those who could benefit from therapies to assist with adjustment and maximise recovery.

Mild traumatic brain injury (mTBI) is a major health problem in the USA. With an incidence of approximately 56.4 per 100 000 people, mTBI accounts for the majority of overall cases of TBI.[1] In recent years, mTBI has gained increasing publicity owing to the reportedly high incidence in soldiers serving in the Operation Enduring Freedom (OEF) and Operation Iraqi Freedom (OIF) conflicts.[2] While these incidence estimates are questionable owing to inconsistencies in defining and screening criteria, there is no doubt that mTBI is a major health problem.

Fortunately, there is evidence that persons with uncomplicated mTBI typically return to preinjury functioning by about 3 months post-injury.[3] Uncomplicated mTBI is defined as an initial Glasgow Coma Scale (GCS) score of 13 to 15, with no evidence of abnormalities on neuroimaging studies or neurological examination.[4] Persons with complicated mTBI (initial GCS score of 13 to 15, but with abnormalities on neuroimaging or neurological examination) typically show a more prolonged course of recovery similar to that of persons with moderate TBI.[4 5] However, a subset of persons with uncomplicated mTBI have symptoms persisting beyond the expected recovery period.[6 7] Symptom persistence has been shown to be positively associated with a history of psychiatric problems, greater emotional reactivity immediately following injury, poor social support and potential secondary gain.[8] Depression may particularly predispose persons with mTBI to show increased symptoms. A recent study found that persons with mTBI who met the DSM-IV criteria for Major Depression endorsed more postconcussive symptoms compared with persons with Major Depressive Disorder alone, persons with mTBI who were not depressed and healthy controls.[9]

Many of the symptoms commonly associated with TBI overlap with those of other disorders, or occur frequently in persons who do not have TBI. These symptoms include changes in physical (eg, headache, fatigue, sleep disturbance), cognitive (eg, memory and concentration difficulties), and emotional (eg, irritability, depression, apathy) functioning. Indeed, Boake and colleagues found that such symptoms were not specific to TBI, but were also reported at a similar rate in a non-brain-injured trauma sample.[10 11] In a civilian sample, symptoms of mTBI have been shown to overlap with symptoms of depression and Post-Traumatic Stress Disorder (PTSD).[8] In a military sample, Hoge *et al* found that the association between mTBI and physical health outcomes disappeared after accounting for depression and PTSD.[12] The overlap between symptom profiles can make it difficult to determine whether persisting symptoms are due to mTBI or are signs of other problems that may or may not be related to the mTBI.

Preinjury stress is a factor that has been minimally considered when predicting outcomes following mTBI. Cumulative environmental stressors are known to have a negative impact on mental health.[13]

Preinjury stress may complicate recovery from mTBI, possibly predisposing persons with injury to develop depression, PTSD and/or other emotional, physical or cognitive symptoms in response to injury. Similarly, the stress associated with a mTBI could exacerbate pre-existing symptoms of depression and/or anxiety. TBI occurs more frequently among individuals of low socio-economic status[14] and in persons with histories of substance use or abuse,[15] increasing the likelihood of preinjury stress. The potential for preinjury stressful life events may be even greater in minorities, as many have a background of socio-economic hardship, which has been associated with poor physical health, impairments in cognitive and social functioning, and emotional distress.[16 17] In support of this, there is a higher rate of violence as a cause of injury in African–American and Hispanics,[18 19] suggesting that they may have been exposed to more stressful life events prior to injury. Understanding the influence of preinjury stressful life events and emotional distress on outcomes following mTBI can help to identify individuals who are at risk for poor outcome and can help target interventions.

The purpose of the current paper is to investigate the incidence of major stressful life events in a sample of persons with medically documented mTBI who were admitted to the emergency centre (EC) of a large, urban Level I trauma centre. The relationship of preinjury stressful life events and post-traumatic stress symptoms to health-related quality of life and emotional distress at 3 months postinjury was also investigated.

METHODS

Study design and participants

The protocol and procedures of the present study were approved by the Institutional Review Board at the Baylor College of Medicine and the Harris County Hospital District, and the Committee for the Protection of Human Subjects at the University of Houston. All participants for this study were recruited from consecutive admissions during specific time intervals to the EC of Ben Taub General Hospital, a Level I trauma centre in Houston, Texas, from April 2000 to January 2004. For inclusion in the present study, a participant must have sustained a mTBI as defined by the Brain Injury Interdisciplinary Special Interest Group of the American Congress of Rehabilitation Medicine.[20] This definition specifies that mTBI is characterised by a traumatically induced physiological disruption of brain function, as manifested by at least one of the following: (1) any period of loss of consciousness (LOC); (2) any loss of memory for events immediately before or after the accident; (3) any alteration in mental status at the time of the accident (eg, disorientation); or (4) focal neurological deficits that may or may not be transient. Medical documentation of altered consciousness was required, as well as an initial postresuscitation GCS of 13–15 in the EC. For cases in which the initial GCS score was temporarily depressed by alcohol, anaesthetics/sedation or intubation, the GCS score upon emergence from the intoxicated or sedated state was used, up to 6 h following injury. Duration of LOC was not documented for the majority of participants. Utilising the injury classification scheme identified by Williams *et al*,[4] participants with both uncomplicated (ie, negative neurological exam and neuroimaging results) and complicated mTBI (ie, either/both positive neurological exam or neuroimaging results) were included. All participants were 18 years of age or greater at the time of consent.

Exclusion criteria included a history or diagnosis of the following conditions: a previous head injury requiring hospitalisation, a central nervous system neurological disorder affecting cognitive functioning of the patient (eg, stroke, dementia, epilepsy) or a major psychiatric disorder (eg, schizophrenia or bipolar disorder). Homeless individuals or individuals visiting the Houston area were excluded because of anticipated follow-up difficulties.

Of 271 persons meeting eligibility criteria, 222 provided informed consent and completed a baseline assessment within 2 weeks of injury. Informed consent and assessment were conducted only once participants were adequately oriented, as defined by a score of 76 or greater on the Galveston Orientation and Amnesia Test.[21] Participants who were not oriented prior to discharge home from the EC provided assent to be contacted within 2 weeks. Those participants admitted to the hospital were approached for participation during their hospital stay. Outcome measures were administered at 3 months postinjury. All measures were administered orally to the participant by an English- or Spanish-speaking research assistant. When Spanish versions of tests were unavailable, the English-language version was translated using a back-translation procedure.

Assessment instruments

Measures of preinjury functioning

For the Stressful Life Events Questionnaire (SLESQ), participants responded in regard to lifetime history, as per the original instructions for the measure. For all other measures, participants were asked to respond to questions according to symptoms experienced during the month prior to injury.

Center for Epidemiological Studies-Depression Scale (CES-D)[22]

The CES-D is a 20-item self-report scale designed to assess depressive symptoms in the general population. Items were selected from a pool of previously validated depression scales. Respondents indicate whether symptoms have been experienced 'rarely or none of the time' (0), 'some or a little of the time,'[1] 'occasionally or a moderate amount of time' (2) or 'most or all of the time' (3) in the past week, yielding total scores ranging from 0 to 60. The internal consistency of the CES-D has been estimated at 0.85 in the general population, and at 0.90 in a psychiatric patient sample.[22] Reliability, validity and factor structure were found to be comparable across a wide variety of demographic characteristics, and scores on the CES-D have moderate to high correlations with other self-report measures of depression. The total CES-D score was used in analyses.

Stressful Life Events Questionnaire (SLESQ)[23]

The SLESQ is a comprehensive, 13-item screening measure of lifetime exposure to traumatic events. The measure is intended to identify exposure to traumatic events that are consistent with the threshold set in Criterion A1 of the PTSD diagnosis in the Diagnostic and Statistical Manual of Mental Disorders-IV (DSM-IV).[24] Like the DSM-IV, traumatic events included in the SLESQ involved 'actual or threatened death or serious injury, or a threat to the physical integrity of self or others' (p. 427, DSM-IV).[24] The SLESQ has been shown to have adequate test–retest reliability and validity.[23] Methods of scoring the SLESQ include total incidence and total frequency.[25] The total incidence and total frequency scores were significantly correlated in this sample ($r=0.876$, $p<0.001$). The total incidence score was used in the current analyses, owing to concerns with accuracy of frequency reporting.

PTSD Checklist (PCL)[26]

The PCL is a 17-item, self-report inventory that assesses symptoms of PTSD. There are three versions of the PCL: military, civilian and specific. The civilian form was used in this study. Respondents are asked to rate the degree to which they have been bothered by a PTSD symptom during the past month (during the month prior to the accident in our sample). Responses range from 1 to 5, with 1 indicating 'not at all' and 5 indicating 'extremely.' The PCL has been shown to have excellent internal consistency, test–retest reliability and validity across diverse samples, including individuals involved in motor vehicle accidents, sexual assault, Vietnam and Persian Gulf veterans, and college students.[27–29] The total score was used in the analyses of this study.

Outcome measures

Thirty-Six-Item Short-Form Health Survey[30] (SF-36) physical and mental health component scores

The SF-36 is a 36-item measure of overall perceived health, with higher scores reflective of more positive health perceptions. Physical and mental health component scores were used in this study. Items pertain to the subjective assessment of functioning, well-being, disability and evaluation of general health. In US general population studies, internal consistency and test–retest estimates range from 0.89 to 0.94 for the physical health component score and from 0.84 to 0.91 for the mental health component scores.[31]

Brief Symptom Inventory (BSI)[32] depression and anxiety subscale scores

The BSI is a 53-item questionnaire requiring individuals to rate the extent to which they have experienced symptoms during the past week. The degree of distress caused by each symptom is rated on a scale ranging from 0 ('not at all') to 4 ('extremely'). The depression and anxiety subscales scores served as outcome variables for the current study. Adequate internal consistency has been demonstrated by Cronbach α coefficients of 0.85 for the depression subscale and 0.81 for the anxiety subscale.[33] Good convergent validity has been demonstrated by maximum correlations between BSI depression and anxiety subscale scores, and corresponding clinical scales of the Minnesota Multiphasic Personality Inventory.[33]

Statistical analysis

Pearson χ^2 were used to investigate racial/ethnic differences in the report of incidence of traumatic events, as assessed by the SLESQ. A series of hierarchical multiple regression analyses were used to investigate the unique contribution of preinjury stressful life events and PTSD symptoms to perceived mental and physical health status at 3 months postinjury after controlling for age, the presence of a CT finding and preinjury depressive symptoms. Age was entered into the first block, presence or absence of a CT finding (coded dichotomously) was entered into the second block, preinjury CES-D total score was entered into the third block, incidence of exposure to traumatic events, as assessed by the SLESQ, was entered into the fourth block, and the PCL total score was entered into the fifth and final block. This regression model was run for each of the outcome variables: BSI depression subscale score, BSI anxiety subscale score, SF-36 mental health component score, and SF-36 physical health component score.

RESULTS

Of the 222 participants who completed the baseline assessment, 186 (84%) completed the 3-month outcome assessment. There were no significant differences between participants who did (N=186) and did not (N=36) complete the outcome assessment with respect to age, education, gender, race, presence or absence of findings on neuroimaging, and scores on preinjury measures, as shown by independent samples t tests and Pearson χ^2. The demographic characteristics of the final study sample are shown in table 1. Over half of the sample was Hispanic, and almost one-quarter was African–American. The sample was evenly divided between persons classified as uncomplicated or complicated mild injury. The ratio of males to females was about 3:1. The majority had less than a high school education and an annual household income of <\$30 000.

Incidence of stressful life events

Table 2 depicts the incidence of stressful life events as assessed by the SLESQ. Pearson χ^2 analyses showed that there were no significant differences in the rate of incidence of the stressful life events across the three racial/ethnic groups. The most frequently occurring stressful life events, endorsed by 25% or more of participants, included: experiencing a life-threatening accident; robbed using physical force or a weapon; the death of an immediate family member/romantic partner/very close friend as the result of an accident/homicide/suicide; and/or witnessing a murder or physical or sexual assault.

Relationship of preinjury stressful life events and post-traumatic stress to emotional distress and health-related quality of life following mTBI

The predictive values for each variable of interest, as well as the covariates, are shown in table 3. After controlling for age, presence of a CT finding and preinjury depressive symptoms, the incidence of stressful life events, as assessed by the SLESQ, was a significant predictor of BSI depression and anxiety subscale scores, and SF-36 physical and mental health component scores. In all cases, a greater number of preinjury stressful life events was associated with worse outcomes. The PCL total score, which was the last independent variable entered in the regression analyses, was a significant predictor of the SF-36 mental health component score only. A greater number of preinjury symptoms of post-traumatic stress disorder was associated with worse health-related quality of life. There was a trend towards significance for the PCL total scores predicting BSI anxiety subscale scores.

Of the covariates, preinjury depression, as assessed by CES-D total score, was a significant predictor of the SF-36 mental

Table 1 Demographic characteristics of study sample

Age, mean (SD, range)	33.63 (12.34, 18–75)
Male, n (%)	143 (77%)
Race/ethnicity, n (%)	
White	28 (15%)
Black	44 (24%)
Hispanic	114 (61%)
Education in years, mean (SD, range)	10.60 (3.95, 0–20)
Annual household income, n (%)	
<\$10 000	32 (17%)
\$10 000–19 999	48 (26%)
\$20 000–29 999	35 (19%)
\$30 000–39 999	16 (9%)
\$40 000–49 999	10 (5%)
\$50 000–74 999	12 (7%)
\$75 000–100 000	6 (3%)
>\$100 000	3 (1%)
Unknown	24 (13%)
Abnormality on head CT, n (%)	90 (49%)

 J Neurol Neurosurg Psychiatry 2011;82:782–787. doi:10.1136/jnnp.2010.228254

Table 2 Incidence of stressful life events on the Stressful Life Events Questionnaire

Stressful life event	Overall incidence	Hispanic incidence	African–American incidence	White incidence
Life-threatening illness	35 (19%)	23 (20%)	7 (16%)	5 (19%)
Life-threatening accident	66 (36%)	34 (30%)	22 (50%)	10 (39%)
Robbed using physical force or a weapon	50 (27%)	30 (27%)	14 (32%)	6 (23%)
Death of immediate family member/romantic partner/very close friend as a result of accident/homicide/suicide	55 (30%)	36 (32%)	9 (21%)	10 (39%)
Raped	11 (6%)	5 (4%)	4 (9%)	2 (8%)
Sexual assault attempted using threat or physical force	7 (4%)	3 (3%)	2 (5%)	2 (8%)
Unwanted touching of private parts when helpless	14 (8%)	6 (5%)	3 (7%)	5 (19%)
Physically abused/attacked/harmed during childhood	34 (19%)	24 (21%)	6 (14%)	4 (15%)
Physically abused/attacked/harmed as an adult	37 (20%)	24 (21%)	7 (19%)	6 (23%)
Threatened with a weapon	31 (17%)	12 (11%)	11 (25%)	8 (31%)
Witnessed a murder or a physical or sexual assault	45 (25%)	26 (23%)	13 (30%)	6 (23%)
Injured/felt life in danger during combat or while living in a war zone	20 (11%)	10 (9%)	5 (11%)	5 (19%)

health component score, the BSI depression subscale score and the BSI anxiety scale. A greater number of preinjury depressive symptoms was associated with worse health-related quality of life and with greater depression and anxiety following injury. Injury severity was predictive of scores on the SF-36 mental health component. Persons with complicated mTBI reported a lower health-related quality of life. Injury severity was not significantly related to scores on the SF-36 physical health component or on the BSI depression and anxiety subscales. Age was not related to outcome for any of the measures.

DISCUSSION

In this sample of people with mTBI, from primarily low socioeconomic and low education backgrounds, potentially life-altering preinjury stressors were common. The most frequently experienced stressful life events included the witnessing of physical or sexual assault, witnessing the death of a loved one by violent or traumatic means, being robbed using physical force or a weapon and experiencing a life-threatening accident. While the history of preinjury stressful life events has not been previously reported for persons with mTBI, the current findings are conceptually consistent with research showing a higher incidence of violent injury in African–Americans and Hispanics.[18 19] While there were no racial/ethnic differences in stressful life events, the sample as a whole was primarily minority and low SES, with the possibility that they had greater exposure to violence compared with the general population. The results may also be conceptually consistent with the research showing that greater economic hardship is associated with greater emotional distress.[16 17] An environment fraught with violence and economic hardship may be associated with a greater incidence of stressful life events, which may predispose to worse outcome following mTBI.

This study was innovative in its emphasis on the role of preinjury stressful life events as a predictor of outcome following mTBI. While previous investigations have shown that a history of emotional difficulties and poor social support are associated with poor outcome following mTBI,[8 9] these investigations have not considered the role of preinjury stress. The current study provides evidence that the history of stressful life events was a significant predictor of emotional distress and health-related quality of life at 3 months postinjury.

The current results indicate that clinicians should consider preinjury stressful life events in treatment planning. Identification of those with pre-existing life traumas through use of a structured interview can help target individuals in need of treatment early after mTBI. As Ruff suggests, regardless of whether the causes of poor outcome following mTBI are neurogenic or psychogenic in nature, a patient-oriented approach would require that appropriate referrals to treatment be made.[34] Ruff also notes that given the heterogeneity of presentation for persons with persisting problems after mTBI, it is likely that a range of treatment modalities could be developed. Cognitive-behavioural treatments (CBT) may hold particular promise for use with clients with cognitive deficits, such as those with mTBI. The inherent structure of such a therapeutic approach, its flexibility and possible use of supplementary written materials may minimise organisational and memory demands.[35 36] In the use of any of these therapeutic approaches, the cultural context must always be considered, particularly given the diversity of backgrounds that exist in this population.

Table 3 Results of regression models predicting Brief Symptom Inventory (BSI) depression and anxiety subscale scores, and 36-item Short-Form Health Survey (SF-36) mental and physical component scores

		BSI-depression subscale score			BSI-anxiety subscale score			SF-36 physical health component score			SF-36 mental health component score		
Block	Variable entered	R^2 change	df	p Value	R^2 change	df	p Value	R^2 change	df	p Value	R^2 change	df	p Value
1	Age	<0.001	1, 173	0.836	0.002	1, 173	0.591	0.003	1, 174	0.472	0.010	1, 174	0.197
2	CT	0.014	1, 172	0.121	0.013	1, 172	0.137	0.025	1, 173	0.038	<0.001	1, 173	0.829
3	Center for Epidemiological Studies-Depression Scale	0.073	1, 171	<0.001*	0.043	1, 171	0.006*	0.038	1, 172	0.009*	0.004	1, 172	0.410
4	Stressful Life Events Questionnaire	0.045	1, 170	0.003*	0.041	1, 170	0.006*	0.067	1, 171	<0.001*	0.028	1, 171	0.026*
5	Post-Traumatic Stress Disorder Checklist	0.011	1, 169	0.145	0.019	1, 169	0.060	0.021	1, 170	0.041*	0.012	1, 170	0.149

*Statistically significant at $p<0.05$.

CBT that are trauma-focused, CBT-based coping skills training and stress management training may be particularly useful in preventing poor outcomes for those with mTBI and a history of other stressful life events. Trauma-focused CBT involves repeated exposure to the trauma memory through writing of a trauma narrative or imagining the traumatic event, in vivo exposure to situations associated with the traumatic event that either cause anxiety or are avoided, and challenging trauma-related appraisals that are maladaptive through behavioural experiments and Socratic questioning.[37] Stress-management training involves training patients in skills such as deep breathing, relaxation skills, positive self-talk, imagery and assertiveness to help with the general management of anxiety.[37] Coping skills training is a more loosely defined intervention that may involve training in a structured problem-solving technique, recognition and changing of maladaptive thinking, pleasant events scheduling and relaxation training.[35]

Findings of this investigation are pertinent to troops returning from the OEF and OIF conflicts. Many of these troops screen positive for possible TBI based on exposure to blast or trauma and self-report of postconcussive symptoms. However, postconcussive symptoms overlap with symptoms of stress disorders that can also occur with high incidence in combat troops. A substantial proportion of troops who screen positive for possible TBI also present with symptoms consistent with depression and/or PTSD.[12] The current findings indicate that emotional adjustment and exposure to stressors prior to deployment may affect symptoms experienced by troops after involvement in combat. The possibility that this cumulative model of stressors operates in OEF/OIF veterans has been proposed by others.[38] In addition, mTBI that occurs during a period of extreme stress may temporarily interfere with normal psychological coping mechanisms resulting in an enhanced stress response. In support of this, Vanderploeg and colleagues have found that persons with PTSD who subsequently experience mTBI have a prolonged recovery from PTSD.[39] Our findings suggest that knowledge of troops' history of lifetime stressful events and emotional distress may permit prediction of response to mTBI incurred in combat. Knowledge of enhanced risk for poor outcome after mTBI might have implications for predeployment interventions to inoculate troops against poor outcome and/or for early interventions after injury to minimise long-term effects.

Over 70 years ago, Sir Charles Symonds reported that '…it is not only the kind of injury that matters, but the kind of head.'[40] Our findings support this observation. 'The kind of head' as operationalised as preinjury exposure to stress and preinjury emotional functioning predicted each of the four indices of psychological outcome after mTBI. In contrast, 'the kind of injury' as operationalised as uncomplicated mTBI as compared with complicated mTBI only predicted the mental health component of the SF-36. Of course, using presence or absence of lesions on initial CT scan is a crude way of measuring 'kind of injury.' It is possible that more sophisticated indices of injury type such as newer MRI sequences may be more predictive of post-mTBI functioning. Future research should employ more sophisticated 'kind of head' and 'kind of injury' variables and search for possible interactions between these patient characteristics.

Acknowledgements We wish to thank the patients and staff of Harris County Hospital District's Ben Taub General Hospital at the time of this study, particularly C Robertson, S Gopinath and A Valadka. Thanks are also warranted to the research assistants who worked on this project, including A Ingraham, N Valles, L Gonzalez and D Mazzei. We would also like to acknowledge WM High, Jr who served as the principal investigator for the overall grant for which these data were collected.

Portions of these data were presented at the meeting of the International Neuropsychological Society, February, 2008, Kona Hawaii and at the Mild TBI 2009—An International Conference on Mild Traumatic Brain Injury, August, 2009, Vancouver, British Columbia.

Funding This work was supported by grants from the National Institute on Disability and Rehabilitation Research, US Department of Education (grant nos H133B990014, H133B090023 and H133A070043).

Competing interests None.

Ethics approval Ethics approval was provided by the Baylor College of Medicine.

Provenance and peer review Not commissioned; externally peer reviewed.

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J Neurol Neurosurg Psychiatry 2011 82: 782-787 originally published online January 17, 2011
doi: 10.1136/jnnp.2010.228254